Wedge geography of erasable and irreducible magic in holographic codes

Ariadne Vale, Tobin Masala, and agy-science
Executable Positive Geometry
epg@subnet345.com

27 July 2026

Abstract

Magic (non-stabilizerness) has been proposed as the microscopic resource behind gravitational back-reaction, but where magic *lives* across the entanglement wedges of an explicit holographic code has not been mapped; the only hardware demonstration of non-local magic to date used two bare qubits. We map it, exactly, in the [[5,1,3]] tile and a six-tile, 20-qubit HaPPY network. Three results. (1) Magic follows *wedge geometry, not region size*: a 12-qubit majority arc that misses the wedge of a magic bulk qubit carries exactly zero magic, while the wedge reads the full injected value $\log_2(4/3)$. (2) For a two-qubit logical state the magic splits exactly into two species — *erasable* (removable by local logical unitaries; isolated operationally by a stabilizer-twin control) and *irreducible* (the non-local magic $M_2^{\mathrm{NL}} = -\log_2(4P^2 - 6P + 3)$ of the reduction purity P, vanishing at Bell and product states and maximal, $\log_2(4/3)$, at $P = 3/4$) — and the decomposition $M_2 = \text{erasable} + M_2^{\mathrm{NL}}$ holds exactly across a full sweep. (3) The two species have *complementary wedge geography*: the erasable component is invisible to every single wedge and appears only in their union, while the irreducible component is readable from *any single wedge* — even a three-qubit sub-wedge — through a junk-corrected boundary purity, with constant correction factors (0.5, 0.25) calibrated once on a stabilizer state. In slogan form: *what local operations can remove, single wedges cannot see; what they cannot remove, single wedges cannot miss.* The one-wedge purity readout is a hardware-ready protocol; we include the honest noise analysis showing the current encoder is noise-dominated on present devices, with a pre-registered lower-bound gate that any redesigned circuit must clear before quota is spent. All results are exact statevector computations for *pure* logical states, cross-checked by three independent agents.

1 Setup

The [[5,1,3]] code encodes one logical qubit in five physical qubits and tiles the HaPPY holographic code [1]; our network is the six-tile, 20-qubit "flower" (one central tile, five petals) used in our earlier wedge and hardware studies. Entanglement-wedge reconstruction in these codes is quantized: a boundary region either contains a bulk qubit's wedge (and can reconstruct it fully) or it does not (and knows nothing).

As the magic measure we use the stabilizer Rényi entropy at $\alpha = 2$ [2]: for a k-qubit density matrix ρ,

$$M_2(\rho) = -\log_2 \left[\frac{\sum_P \mathrm{Tr}(\rho P)^4}{\sum_P \mathrm{Tr}(\rho P)^2} \right], \tag{1}$$

the sum over all 4^k Pauli strings, computed exactly via fast Walsh–Hadamard transforms of the state's Pauli spectrum (engine self-tested against direct enumeration). M_2 vanishes on stabilizer states, pure or mixed. *Stated caveat:* for mixed reductions M_2 is a computable magic diagnostic, not a proven monotone; the twin-control construction below exists precisely because mixed-state M_2 alone false-positives on classical mixtures of stabilizer states.

2 Magic follows the wedge, not the region

Injecting a T-state ($|T\rangle \propto |0\rangle + e^{i\pi/4}|1\rangle$) into a bulk slot and sweeping boundary regions: a region containing the bulk qubit's wedge reads the full injected magic $\log_2(4/3) \approx 0.415$; a **12-qubit majority arc that misses the wedge reads exactly zero**. Control injections (stabilizer bulk states) read zero everywhere. Magic does not leak, dilute, or spread with region size — it is wedge-quantized, exactly as reconstructable information is. This was a pre-registered prediction (the wedge reduction inherits the logical Pauli spectrum intact), confirmed by the sweep.

3 Two species of magic, exactly decomposed

For a two-qubit logical state $|\psi_L\rangle$ shared between two tiles, the literature's non-local magic [3] is the minimum SRE over local unitaries, with closed form

$$M_2^{\mathrm{NL}}(P) = -\log_2\big(4P^2 - 6P + 3\big), \qquad P = \operatorname{Tr}\rho_1^2, \tag{2}$$

where ρ_1 is the one-logical-qubit reduction. M_2^{NL} vanishes at $P = 1/2$ (Bell) and $P = 1$ (product) and attains its maximum $\log_2(4/3)$ at $P = 3/4$; it is *not* monotone in P.[1] We define the *erasable* component as the remainder $M_2 - M_2^{\mathrm{NL}}$, and isolate it operationally with the *stabilizer-twin control*: a phase-stripped twin state with identical mixedness whose subtraction removes amplitude-magic common to both.

Sweeping the logical family $|\psi_L(t)\rangle = \cos t\,|00\rangle + e^{i\pi/4}\sin t\,|11\rangle$ (exact, on the code):

t/π	M_2 (total)	$M_2^{\mathrm{NL}}(P)$	direct min-SRE	erasable = $M_2 - M_2^{\mathrm{NL}}$	twin-Δ
0.083	0.3276	0.2996	0.2996	0.0280	0.0280
0.125	0.5406	0.4150	0.4150	0.1255	0.1255
0.167	0.5737	0.2996	0.2996	0.2742	0.2742
0.250	0.4150	0.0000	0.0000	0.4150	0.4150
0.333	0.5737	0.2996	0.2996	0.2742	0.2742

Both identities hold to machine precision across the sweep: the closed form equals direct numerical minimization over local unitaries (validating Eq. (2) on the code), and the erasable remainder equals the twin-control extraction. At $t = \pi/4$ (Bell) *all* magic is erasable; at $t = \pi/8$ the irreducible component is maximal.

4 Complementary wedge geography

Where does each species live on the 20-qubit network? The erasable component is invisible to every single wedge and appears only in the union of both wedges (established in our Phase-1 sweep: single-wedge twin-Δ reads zero across the family). The irreducible component is the opposite: because

[1] This behavior — zeros at both ends, peak at $P = 3/4$ — was independently re-derived from Eq. (2) by the analytical reviewer during this project's audit round, catching a mis-statement in an internal communication; the labs' recorded results were unaffected.

purity carries the Schmidt spectrum, and any wedge reconstructs its logical qubit's reduction up to a *constant* junk factor, M_2^{NL} is readable from any single wedge by junk-corrected purity:

$$P \;=\; \mathrm{Tr}\,\rho_W^2\,/\,c_W, \tag{3}$$

with c_W calibrated once on a stabilizer (product) logical state: $c_W = 0.5$ for a 4-qubit wedge, $c_W = 0.25$ for a 3-qubit sub-wedge. Exactly, across the sweep:

t/π	M_2^{NL} true	from 4q wedge	from 3q sub-wedge	erasable in one wedge
0.083	0.2996	0.2996	0.2996	0.0000
0.125	0.4150	0.4150	0.4150	0.0000
0.167	0.2996	0.2996	0.2996	0.0000
0.250	0.0000	0.0000	0.0000	0.0000
0.333	0.2996	0.2996	0.2996	0.0000

In slogan form: **what local operations can remove, single wedges cannot see; what they cannot remove, single wedges cannot miss.** The mechanism is elementary once seen — reductions carry Schmidt data (irreducible), relative phases do not (erasable) — but the code instantiation is, to our knowledge, the first wedge-resolved map of the two species, and it converts directly into a measurement protocol.

5 The one-wedge hardware protocol, and an honest noise analysis

Because M_2^{NL} needs only a *purity*, it is measurable from a single wedge by randomized measurements — no tomography, no magic witness on the full state. We constructed the exact two-tile encoder (a GF(2) symplectic completion of the stabilizer group); noiseless extraction through the full circuit reproduces $M_2^{\mathrm{NL}} = 0.29956$ exactly at the test point.

We report the negative result with the same care. The registered quantitative basis is a routed rehearsal manifest: every randomized-measurement circuit is transpiled through the target (Heron-class) backend's coupling and layout before backend-derived noise is applied, averaged over five replicates (120 settings $\times$ 512 shots each). It gives point estimates $M_2^{\mathrm{NL}} = 0.165$ (baseline) and 0.183 (best resynthesis candidate) with $\sigma_{\mathrm{total}} \approx 0.098$–$0.100$ — so the lower bound falls below the 0.10 gate at practical shot budgets, and the physical mechanism remains adverse: decoherence mixes the wedge in exactly the channel the signal uses ($P \to 1/2$). (An earlier unrouted noise-floor probe — the noise model applied without backend routing or layout, single seed — read $M_2^{\mathrm{NL}} \approx 0.005$, clipping at the maximally-mixed floor; we retract that figure as an obsolete pipeline bracket, superseded by the routed manifest.) Any hardware attempt is therefore pre-registered against a lower-bound gate: with $\sigma_{\mathrm{total}} = \sigma_{\mathrm{shot}} + \mathrm{SEM}_{\mathrm{rehearsal}}$ (additive, conservatively — the purity estimator's shot error plus the rehearsal ensemble spread), a redesigned circuit must achieve $M_2^{\mathrm{NL}}(\text{rehearsal}) - 2\sigma_{\mathrm{total}} > 0.10$ (point estimate $\gtrsim 0.19$ at practical shot budgets) before device quota is spent; otherwise the deliverable is a quantified noise bound, not a forced signal. The redesign campaign and its outcome will be reported against this gate as registered.

6 Relation to prior work, and honest scope

Non-local magic and its closed form Eq. (2) come from the two-qubit experiment of [3]; the motivation that non-trivial area operators *require* non-local magic, and that back-reaction is magical, is from [4, 5]; state-dependent geometry from magic-enriched codes is [6]. Our contribution is the *wedge-resolved* map in explicit holographic codes — the geography, the exact two-species decomposition on a code, the twin control, and the junk-corrected one-wedge readout as a hardware-ready protocol.

Scope. All results are exact statevector computations for *pure* two-qubit logical states (the analytical reviewer's audit fixed this scoping: the erasable/irreducible split via reduction purity is defined here for pure logical states; the mixed-logical-state extension requires a consistent definition of irreducible magic and is left open). Mixed-state M_2 is used only as a diagnostic, backed by the twin control. No hardware measurement of wedge magic is claimed — the protocol is ready, the current circuit fails the registered gate, and the redesign is in progress.

Methods — three-agent adversarial verification

Built and verified by three AI research agents on distinct runtimes under human direction: exact construction and synthesis (quant-phy/Ariadne, Anthropic), independent execution and noise rehearsal (codex-science/Tobin, OpenAI), and analytical audit (agy-science) — whose review corrected a mis-stated property of Eq. (2) in internal communication and fixed the pure-state scoping above. Every number is reproducible from the public repository (`labs/track-d-magic/`, Phases 0–1b and `species_hardware.py`); the noise-gate criterion is pre-registered in the repository before any device run.

References

[1] F. Pastawski, B. Yoshida, D. Harlow, J. Preskill, *Holographic quantum error-correcting codes*, JHEP **06** (2015) 149, arXiv:1503.06237.

[2] L. Leone, S. F. E. Oliviero, A. Hamma, *Stabilizer Rényi Entropy*, Phys. Rev. Lett. **128**, 050402 (2022), arXiv:2106.12587.

[3] *Experimental demonstration of non-local magic in a superconducting quantum processor*, arXiv:2511.15576.

[4] C. Cao et al., *Non-trivial Area Operators Require Non-local Magic*, JHEP **11** (2024) 105, arXiv:2306.14996.

[5] *Gravitational back-reaction is magical*, arXiv:2403.07056.

[6] C. Cao, W. Cheng, S. Karthikeyan, Y. Li, J. Preskill, *State-dependent geometries from magic-enriched quantum codes*, arXiv:2603.13475.

Acknowledgements. Produced by a swarm of AI research agents — OpenAI, Anthropic, and a third analytical-reviewer agent — under human direction, each independently checking the others before a claim was allowed to stand. Code and data public; every number carries a commit hash.